

Kanishk Yadav

✉ kanishkyadav1010@gmail.com |  LinkedIn |  GitHub |  ky1452.github.io

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

M.S. in Computer Science (AI & ML Specialization)

Jan 2027 – May 2028

Coursework: Distributed Systems, Adv. NLP, ML Systems, Adv. Data Management, Applied Parallel Programming

Birla Institute of Technology and Science, Pilani

Hyderabad, India

B.E. in Computer Science & M.Sc. in Chemistry | CGPA: 8.7/10

Aug 2019 – Jul 2024

Coursework: Data Structures & Algorithms, ML, NLP, CV, OS, Networks, Databases, OOP, Linear Algebra, Prob & Stats

WORK EXPERIENCE

Walmart

Bengaluru, India

Software Engineer III (fast-track promotion), Walmart OMS

June 2026 – Present

- Spearheading the platformization of Sam's Club shipment tracking onto Walmart's multi-tenant enterprise architecture
- Architected a centralized E2E automation framework with BOM and JDK21, reducing production regressions by >30%

Software Engineer II, Sam's Club OMS

Jul 2024 – May 2026

- Modernized global shipment tracking, transitioning from monolith to scalable microservices handling 1M+ tracking events daily
- Designed KubeInSight, a LangGraph multi-agent K8s tool for automated RCCA, reducing incident triage from 60+ to <10 mins
- Migrated cron-based polling to event-driven webhooks for real-time tracking of 95% Walmart orders, increasing GMV by \$31M
- Led org-wide cost optimizations in Azure and Kafka resources across 11 supply chain teams, delivering \$1M in annual savings

Software Engineer Intern

May 2023 – Jul 2023

- Engineered an NLP pipeline (TF-IDF and SVMs), improving milestone-mapping by 75% and eliminating human-in-the-loop
- Built a web app aggregating global shipment data from CosmosDB and OracleDB, reducing manual lookup latency by 90%

Bharat Electronics Limited

Bengaluru, India

Deep Learning Intern [Report]

May 2021 – Jul 2021

- Trained YOLOv5 for the Indian Coast Guard to automate surveillance of 7,500 km of Indian coastline, achieving 0.86 mAP@0.5
- Achieved 4x faster training convergence through tuning; deployed to detect 9 maritime classes, analyzed OOD gaps in field tests

RESEARCH EXPERIENCE

Nanyang Technological University

Singapore

Undergraduate Researcher | Advisor: Prof. Chng Eng Siong [Code] [Thesis]

Jan 2024 – Jul 2024

- Achieved 87% accuracy and a 33% improvement in F1-score for speaker counting on the DIHARD III dataset; engineered a sequential fine-tuning pipeline using WavLM to resolve class imbalance
- Benchmarked performance-latency tradeoffs in Pyannote's Voice Activity Detection for low-latency real-time streaming

University of Toronto

Toronto, Canada

Mitacs Globalink Scholar | Advisor: Prof. Oleksandr Voznyy [Code] [Thesis] [Publication]

Jul 2023 – Dec 2023

- Architected an ML testbed for band gap prediction to capture individual material properties from discrete atomic embeddings
- Designed shared-weight Siamese-like network architectures with permutation invariance to eliminate positional bias outliers
- Validated robustness and generalizability in extreme low-data (1% train split) regimes by scaling from 5-atom to 60-atom tasks

AWARDS & RECOGNITION

- **Bravo Award, Walmart:** Recognized at the org-level for delivering key cost-optimization and modernization milestones
- **Making A Difference Award, Walmart:** Awarded for completing critical platformization initiatives for Walmart markets
- **Mitacs Globalink Scholarship:** Awarded 9,200 CAD for funded research at UofT; selected from 17,000+ applicants globally

SELECTED PROJECTS

ErrorCortex: RAG-Powered Error Intelligence [Code]

Oct 2025 – Present

- Architected an agentic SDK optimizing SDLC by intercepting, offloading runtime exceptions via RabbitMQ for RAG reasoning
- Synthesized code patches using Llama-3 grounded by heuristic context and enforced syntactic validity via AST verification
- Designed a self-improving feedback loop and re-indexed human-approved fixes into ChromaDB to refine future predictions

Conversational AI Chatbot | Chubb Insurance [Report]

Jan 2023 – Jun 2023

- Developed a chatbot handling both FAQ and transactional queries by building and querying a Neo4j Knowledge Graph
- Engineered few-shot intent classification using SetFit and SBERT semantic fallback for high accuracy with limited data

TECHNICAL SKILLS

Languages: Python, Java, C++, C, Go, SQL, L^AT_EX

AI/ML Frameworks & Tools: PyTorch, TensorFlow, CUDA, Claude Code, Copilot, scikit-learn, LangChain, NumPy, Pandas

Systems & Cloud: Docker, Kubernetes, Kafka, RabbitMQ, Postgres, Elasticsearch, Redis, Splunk, Azure, Grafana, Prometheus, Git, REST APIs, CI/CD, Swagger, Postman